When the control record arrives: a finite-error memory law for one quantum payload

Constructed research return with independent audits

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Abstract

A stream may contain only one arbitrary qubit of recoverable payload yet require extensive quantum memory before a short classical control record arrives. We give a quantitative theorem for this task, allowing unlimited classical records, arbitrary receiver processing and a different final decoder for every record. If balanced encoding observables have commutator gap κ, a receiver with quantum dimension at most m in each classical branch and selector-averaged half-diamond error $\epsilon \le 1/2$ must satisfy $m \ge N[1 - 8\epsilon(1-\epsilon)/\kappa]$, where N is the early register's dimension. A common Petz reconstruction is constructed only as a proof witness; whole-output reconstruction is not required of the receiver.

A small-bias Pauli construction gives $\kappa \ge 4/5$, a once-through tracial stream of n carriers and one late payload flag, a $2\lceil \log_2(4n-2) \rceil + 1$ bit record, and one coherent source qubit. Delivering that record before the carriers permits exact arbitrary-reference recovery by local measurements, one classical parity and a final one-qubit correction: linear quantum work, constant coherent receiver workspace and near-linear classical work. At one percent error, delaying the same record requires $m \ge 0.901\, 2^{n+1}$, so a receiver must provision all $n+1$ whole qubits of capacity at the critical cut; individual classical branches may retain fewer. A separate imported local expander gives a five-bit record and constant total logical source dimension, with a much smaller certified error threshold. These are complete programmed-source constructions, not a decoder for the homogeneous weak partial-SWAP stream. Strong encounters, paid source coherence, the actual timing of the control record and implementation accuracy are part of the statements.

1 The selected question and its consequence

The completed fourth-order action law determines how much mixed-particle interaction is necessary for physical output acquisition. It does not determine how much quantum memory a receiver must keep. The next question selected here is:

> Can the dimension of a small recoverable payload, or of a small emitting source, bound the quantum memory of a causal receiver when its physical inputs are mixed? Which additional classical access makes a small receiver possible?

The answer is a quantitative obstruction with a matching positive interface. The relevant distinction is the arrival time of a record specifying the source's encoding. A short early record can enable immediate local measurements and exact recovery. The same record delivered later can force retention of essentially all quantum carriers, even though the final requested output is one qubit.

The mathematical novelty claimed in this return is limited to the derivation and composition given here. Late-choice quantum-memory phenomena are established [1, 2]; environmental measurement

and correction are established [3]; small-bias Pauli randomization is established [5, 6]. We supply an explicit finite-error bridge under the weaker, payload-specific recovery premise and complete physical stream constructions. No historical priority, laboratory demonstration, or realised civilisation-scale benefit is asserted. The result does rule out a general compact mixed-stream receiver based only on payload dimension, and identifies an access resource that removes the obstruction in the constructed family.

Independent selection initially pursued the weak partial-SWAP receiver, thermal identification from partial observation, raw-image hydrodynamic force acquisition, and this foundational memory question. The latter yielded the strongest closed theorem and operation. The other completed results are retained as companions, with their actual costs and boundaries in Section 10.

2 Operational contract

The payload Q is an arbitrary qubit, possibly entangled with an arbitrary reference R. A classical selector j, independent of QR, specifies a known encoding. The early quantum register B has dimension $N = 2D$ and is

$$\mathcal{E}_j(\rho) = V_j(\rho \otimes I_D/D)V_j^\dagger. \tag{1}$$

All of B arrives before j in the hard interface. The receiver applies one common channel $\mathcal{C}$ before seeing j. Its retained state consists of an arbitrary classical record and a quantum register of dimension at most m in every classical branch. After learning j, it may use any channel $\mathcal{D}_j$ to return one qubit. Fresh decoder ancillas are allowed. There is no uncounted quantum memory, entanglement half, output delay line or coherent external record across the cut. No initial entanglement with the source is supplied.

We use the half-diamond convention throughout:

$$d_\diamond(\mathcal{A}, \mathcal{B}) = \tfrac{1}{2} \left\| \mathcal{A} - \mathcal{B} \right\|_\diamond. \tag{2}$$

The sufficient performance premise is only

$$d_\diamond \left(\sum_j p_j \mathcal{D}_j \mathcal{C} \mathcal{E}_j, \mathrm{id}_Q \right) \leq \epsilon. \tag{3}$$

A worst-selector or flagged reference-uniform guarantee implies this premise. The different $\mathcal{D}_j$ need not reconstruct the full state of B, preserve every mixed-output correlation, or be related to one another.

The bound is on quantum dimension in every classical branch, not expected qubit count. It allows arbitrary adaptive measurements and classical feedback inside the receiver. In fact it allows a collective operation on all of B before the cut, and therefore also bounds every causal once-through receiver. The source follows its declared program; receiver feedback cannot reprogram it to release the payload or selector earlier. Failure/postselection is not omitted from (3).

3 The memory theorem

Let $H_j = V_j(Z \otimes I_D)V_j^\dagger$. These are traceless Hermitian involutions. Assume the following quantitative irreducibility for every complex $N \times N$ matrix K, using unnormalised Hilbert–Schmidt norms:

$$\sum_j p_j \left\| [K, H_j] \right\|_2^2 \geq \kappa \left\| K - \tfrac{\mathrm{Tr}\, K}{N} I_N \right\|_2^2. \tag{4}$$

Theorem 1 (Payload memory with a late selector)**.** *Under* (1)–(4)*, for* $0 \le \epsilon \le 1/2$*,*

$$\boxed{\frac{m}{N} \ge 1 - \frac{8\epsilon(1-\epsilon)}{\kappa}.} \tag{5}$$

The lower bound can be replaced by zero when negative. In particular, $\epsilon \le \kappa/16$ *implies* $m \ge N/2$*. Exact recovery requires* $m \ge N$*.*

Proof. Set $\sigma = \mathcal{C}(I_N/N)$ and $x_j = \mathcal{C}(H_j)/N$. Positivity gives $-\sigma \le x_j \le \sigma$, so $t_j = \|x_j\|_1 \in [0,1]$. The two payload inputs $|0\rangle, |1\rangle$ encode as $(I_N \pm H_j)/N$. Their optimum discrimination bias after compression is t_j. Measuring Z after $\mathcal{D}_j$ is an allowed, possibly suboptimal test. Applying (3) to each of these two inputs shows that its selector-averaged success probability is at least $1-\epsilon$. Consequently

$$\sum_j p_j t_j \ge 1 - 2\epsilon. \tag{6}$$

Construct the Petz map relative to the full-rank input I_N/N:

$$\mathcal{R}(X) = \frac{1}{N}\mathcal{C}^*(\sigma^{-1/2} X \sigma^{-1/2}). \tag{7}$$

Inverses are on $\operatorname{supp}\sigma$. Every output of $\mathcal{C}$ is supported there. The map is completely positive and trace preserving on that support, since $\mathcal{C}(I_N) = N\sigma$; extend it arbitrarily to a trace-preserving map on the unused complement. Thus $\mathcal{T} = \mathcal{R}\mathcal{C}$ is a channel through the same hybrid memory. Define

$$q_j = \frac{1}{N}\operatorname{Tr}[H_j \mathcal{T}(H_j)] = \operatorname{Tr}[x_j \sigma^{-1/2} x_j \sigma^{-1/2}]. \tag{8}$$

For $A_j = \operatorname{sign}(x_j)$, extended as a Hermitian contraction, weighted Cauchy–Schwarz yields

$$t_j^2 \le \operatorname{Tr}[\sigma^{1/2} A_j \sigma^{1/2} A_j]\, q_j \le q_j. \tag{9}$$

For completeness, the first factor is $\sum_{ab} \sqrt{\sigma_a \sigma_b}|(A_j)_{ab}|^2$ in an eigenbasis of σ. Using $\sqrt{uv} \le (u+v)/2$ and $A_j^2 \le I$ bounds it by $\operatorname{Tr}\sigma = 1$.

Write $\mathcal{T}(X) = \sum_\ell K_\ell X K_\ell^\dagger$. Trace preservation and $H_j^2 = I$ give the exact identity

$$\sum_\ell \|[K_\ell, H_j]\|_2^2 = 2N - 2\operatorname{Re}\operatorname{Tr}[H_j \mathcal{T}(H_j)]. \tag{10}$$

Applying (4) to each Kraus operator gives

$$F_e(\mathcal{T}) = N^{-2} \sum_\ell |\operatorname{Tr} K_\ell|^2 \tag{11}$$

$$\ge 1 - \frac{2}{\kappa}\left(1 - \sum_j p_j q_j\right) \tag{12}$$

$$\ge 1 - \frac{2}{\kappa}\left[1 - \left(\sum_j p_j t_j\right)^2\right] \ge 1 - \frac{8\epsilon(1-\epsilon)}{\kappa}. \tag{13}$$

Here Jensen's inequality and (9) supply the middle step.

Every composite Kraus operator of a channel through dimension m in each classical branch has rank at most m. Therefore $|\operatorname{Tr} K_\ell|^2 \le m \operatorname{Tr} K_\ell^\dagger K_\ell$, and $F_e(\mathcal{T}) \le m/N$. Equivalently its Choi state has Schmidt number at most m, whose overlap with an N-dimensional Bell state is at most m/N. This argument is independent of the classical alphabet size; its closed-convex form also covers limits of dephased classical records. Combining with (13) proves the theorem. □

The key step is (9). The actual task allows a different decoder after every selector and returns only a qubit. It does not supply a common whole-output reconstruction. The proof manufactures one analytically from the compression channel. No efficient physical implementation of (7) is assumed or needed.

3.1 Causal prefixes and retained quantum exposure

Suppose B consists of s qubits emitted in a fixed order, with the selector only afterward. Let $\mathcal{C}_\ell$ be the receiver's operation after the first ℓ qubits. Because future source operations do not act on emitted qubits or depend on receiver feedback, $\mathcal{C}_\ell$ can be deferred algebraically to act on the corresponding prefix of the complete emitted state. Granting the receiver the untouched suffix for free gives a valid compression $\mathcal{C}_\ell \otimes \mathrm{id}_{\mathrm{suffix}}$ through dimension at most $m_\ell 2^{s-\ell}$ per classical branch. Theorem 1 implies

$$\frac{m_\ell}{2^\ell} \ge r_\epsilon, \qquad r_\epsilon = 1 - \frac{8\epsilon(1-\epsilon)}{\kappa}. \tag{14}$$

Thus the obstruction occurs throughout the stream, not only at its end. For $r_\epsilon > 1/2$, a provisioned register of whole qubits needs capacity at least ℓ after every prefix.

Variable branch sizes can also be charged without adding maxima from incompatible histories. Let branch y retain $q_y = \log_2 m_y$ qubits of dimension. Under the maximally mixed payload (or the marginal of a Bell input), the prefix baseline is $I/2^\ell$. The branchwise rank proof for the Petz witness gives

$$r_\epsilon \le F_e(\mathcal{T}) \le \mathbb{E}\, 2^{\min(q_y,\ell)-\ell}. \tag{15}$$

Indeed each branch of the Petz reconstruction is trace preserving on that branch's support, so its Kraus squared norms sum to the baseline branch probability times the input dimension. The prefix input additionally bounds each compression Kraus rank by 2^ℓ. Convexity places $2^{x-\ell}$ below the chord connecting $x = 0$ and $x = \ell$, yielding

$$\mathbb{E} q_y \ge \max\left\{0, \frac{\ell(r_\epsilon - 2^{-\ell})}{1 - 2^{-\ell}}\right\}. \tag{16}$$

This expectation is for the stated baseline input; it supplies a worst-input resource lower bound for a reader required to work on arbitrary inputs.

If arrivals are separated by at least τ, apply the same argument at every time between arrivals and integrate. The expected retained qubit-time before the selector is at least

$$\tau \sum_{\ell=1}^{s-1} \max\left\{0, \frac{\ell(r_\epsilon - 2^{-\ell})}{1 - 2^{-\ell}}\right\}. \tag{17}$$

For fixed positive r_ϵ this is $\Omega(\tau s^2)$. At one percent in the small-bias construction, $r_\epsilon = 0.901$. This is an actual mean-exposure lower bound, not a sum of branchwise worst-case costs. Extra waiting before any selector information arrives obeys the same prefix bound. Progressive header disclosure requires a conditional-ensemble analysis. Source feedback that changes the encoding would invalidate the deferred-prefix step and is an extra interface.

3.2 An experimentally scored consequence

Choose j with probability p_j and an independent uniform sign. Prepare $(I_N \pm H_j)/N$, hide the labels during compression, then disclose only j and ask for the sign. If the average error is $q \le 1/2$, the same proof gives

$$\frac{m}{N} \ge 1 - \frac{8q(1-q)}{\kappa} = 1 - \frac{2}{\kappa}(1-s^2), \qquad s = 1-2q. \tag{18}$$

This witnesses quantum memory under trusted preparation; a high score alone does not establish arbitrary-payload recovery. For M independent trials of a fixed device with signed mean $\widehat{s}$ and trusted score bias at most ν, Hoeffding gives

$$s_{\mathrm{low}} = \max\{0, \widehat{s} - \sqrt{2\log(1/\delta)/M} - \nu\} \tag{19}$$

with confidence $1-\delta$. Substitution in (18) gives a finite certificate. The per-trial physical preparation still uses n carriers and $O(n)$ operations. The claim is not device independent, does not exclude unmodelled quantum side channels, and does not count only successful trials.

If all M signed trials succeed and trusted bias is zero, the exact binomial bound is $q \le 1-\delta^{1/M} \le \log(1/\delta)/M$. Therefore $M \ge 8\log(1/\delta)/(\kappa\eta)$ suffices to certify $m/N \ge 1-\eta$. For the small-bias gap $\kappa = 4/5$, $\eta = 0.1$ and confidence 95%, 300 zero-failure trials suffice. This is a calculated ideal qualification point, not an executed 1000-qubit experiment. Nonzero trusted bias must be subtracted. Each trial uses one known pure payload qubit, n tracial carriers, the paid source control and delayed record, and one final bit. Applying the prefix argument to the same score also certifies the mean retained quantum exposure in (17); the test does not certify universal recovery merely because it certifies that the receiver had to preserve quantum memory.

4 A short explicit control record with a useful gap

4.1 A balanced Hermitian lift

Suppose Hermitian involutions U_j on a D-dimensional gauge register have average conjugation Φ with absolute Hilbert–Schmidt contraction at most $1-\gamma$ on traceless matrices. Use

$$H_0 = Z \otimes I_D, \quad H_j = X \otimes U_j, \quad p_0 = \alpha, \quad \sum_{j\neq 0} p_j = 1-\alpha. \tag{20}$$

The conditional distribution on $j \neq 0$ is that of Φ. Then

$$\kappa \ge \min\{4\alpha,\ 2(1-\alpha)\gamma\}. \tag{21}$$

To see this, decompose $K = I \otimes A + X \otimes B + Y \otimes C + Z \otimes E$. Average conjugation by (20) is block diagonal in these four flag sectors. On the gauge-traceless part, the largest possible eigenvalue is at most $1-(1-\alpha)\gamma$. The remaining nonscalar flag modes have gaps 2α, 2 and $2(1-\alpha)$ from eigenvalue one. Commutator energy is twice this spectral deficit, proving (21) for complex K. The Z anchor is essential: $X \otimes I_D$ commutes with every $X \otimes U_j$.

The encoders are

$$V_0 = I, \qquad V_j = (|0\rangle\langle 0| \otimes U_j + |1\rangle\langle 1| \otimes I_D)\,(H \otimes I_D). \tag{22}$$

Here H is Hadamard. Direct multiplication gives $V_j(Z \otimes I_D)V_j^\dagger = X \otimes U_j$.

4.2 Small-bias Pauli construction

The following finite-field construction is standard small-bias machinery [5]; the Pauli connection is established [6]. We include the calculation so the gap does not depend on an unspecified random family.

Let $r = 2n$, $k = \lceil \log_2(2(r-1)) \rceil$, $q = 2^k$, and choose a, b uniformly in $\mathbb{F}_q$. For $0 \le \ell < r$, set

$$x_\ell = \mathrm{tr}_{\mathbb{F}_q/\mathbb{F}_2}(ab^\ell), \qquad b^0 = 1. \tag{23}$$

For every nonzero $v \in \mathbb{F}_2^r$, let $f_v(z) = \sum_{\ell=0}^{r-1} v_\ell z^\ell$. Uniformity of a gives

$$\mathbb{E}_{a,b}(-1)^{v\cdot x} = \Pr_b\{f_v(b) = 0\} \le \frac{r-1}{q} \le \frac{1}{2}. \tag{24}$$

The equality uses nondegeneracy of the field trace; the inequality counts roots of a nonzero polynomial. The biases are also nonnegative.

Pair the bits into n Hermitian one-qubit Paulis,

$$P_{a,b} = \bigotimes_{i=0}^{n-1} i^{x_{2i}x_{2i+1}} X^{x_{2i}} Z^{x_{2i+1}}. \tag{25}$$

Every nonidentity Pauli is an eigenoperator of average conjugation by $P_{a,b}$. Its eigenvalue is a nontrivial binary character average of the form (24). Thus the absolute contraction on traceless matrices is at most $1/2$. In (20) choose $\alpha = 1/5$, obtaining

$$\boxed{\kappa \ge 4/5.} \tag{26}$$

The record is a branch bit plus the two k-bit field elements, padded also in the anchor branch:

$$L = 2\lceil \log_2(4n-2) \rceil + 1 = O(\log n). \tag{27}$$

This is a declared source-program distribution independent of the unknown payload. A worst-record guarantee implies the theorem under these weights. The branch coin and seed require their stated accuracy; they are not bath preparation labels.

Generate the Pauli word online with $O(k)$ classical bits: update $y_\ell = ab^\ell$ by $y_{\ell+1} = by_\ell$ and evaluate a precomputed binary linear trace functional. Elementary polynomial-basis multiplication costs $O(k^2)$ bit operations, giving $O(nk^2)$ work without enumerating q^2 words. A field polynomial can be supplied and verified or found by enumerating $O(q)$ monic candidates and testing irreducibility with modular Frobenius powers and polynomial gcds. A conservative deterministic setup bound is $O(qk^3 + k^4) = O(n \log^3 n + \log^4 n)$. The field definition, program length and clock are classical resources.

4.3 Concrete memory consequences

For $N = 2^{n+1}$, Theorem 1 gives

$$m \ge 2^{n+1}[1 - 10\epsilon(1-\epsilon)]. \tag{28}$$

At $\epsilon = 0.01$ the factor is 0.901. A memory made of whole qubits must have capacity $n+1$ at the cut after the flag and before the record. Immediately before the flag arrives its required capacity is n: the flag can at most double the available memory dimension. Capacity here is the maximum

over classical branches; an individual branch may retain fewer. This is a dimension bound, not a requirement to keep the original carriers without encoding. Compressing away one whole qubit, $m \le N/2$, requires

$$\epsilon \ge \frac{1-\sqrt{0.8}}{2} \simeq 0.0527864. \tag{29}$$

For $n = 1000$, $q = 4096$, $k = 12$ and $L = 25$ bits. One-percent recovery needs capacity for 1001 qubits after the flag, or 1000 before it. The separate expected-memory bound is (16). These counts are algebraic consequences; a 1000-qubit experiment was not run.

Uniform sampling from all 4^n Paulis instead gives $\gamma = 1$ and, with $\alpha = 1/3$, $\kappa = 4/3$. Its factor at one percent is 0.9406, at a $2n+1$ bit record cost. The logarithmic record is the selected quantitative construction, not an assumption of uncharged random global unitaries.

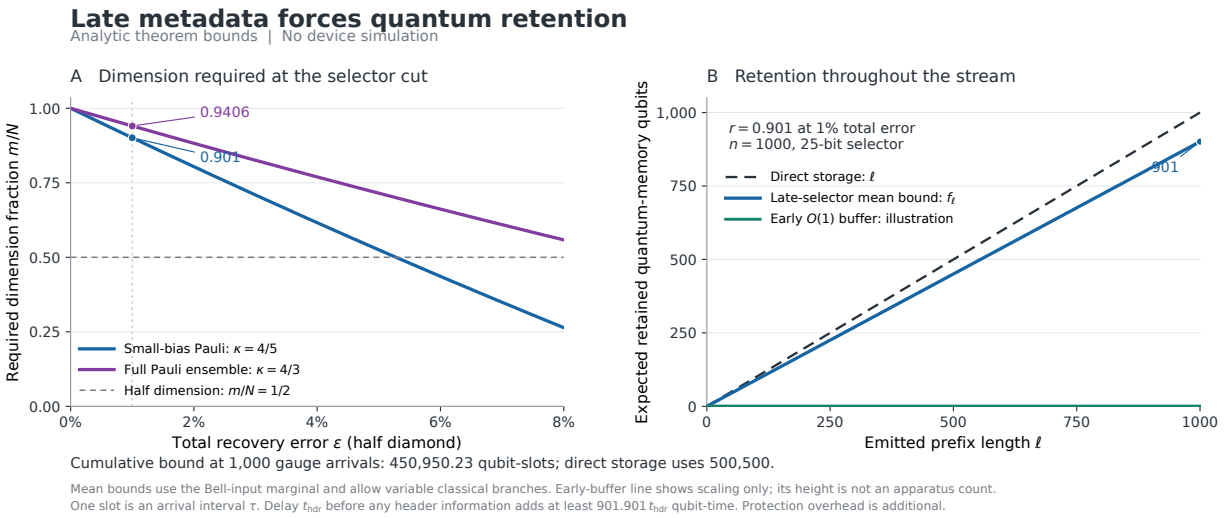


Figure 1: Analytic dimension and mean-retention bounds. The early-buffer line illustrates constant scaling, not an exact apparatus count. With 1000 gauge carriers and one-percent error, the pre-record expected receiver exposure is at least $450{,}950.22834\,\tau$ qubit-time. Direct storage uses $500{,}500\,\tau$ over those intervals. All physical protection is additional.

5 The complete early-record quantum operation

Write U_j as a product of disjoint local factors. Let Π_z refine their joint spectral projectors, with $U_j\Pi_z = u_z\Pi_z$, $|u_z| = 1$. The controlled encoding obeys the exact instrument identity

$$(I \otimes \Pi_z)V_j = [\mathrm{diag}(u_z, 1)H] \otimes \Pi_z. \tag{30}$$

For an arbitrary initial $\rho_{RQ} \otimes \Gamma_{GF}$, where F can be an inaccessible gauge purifier, the unnormalised conditional output factors as

$$(I_R \otimes W_z)\rho_{RQ}(I_R \otimes W_z^\dagger) \ \otimes \ \mathrm{Tr}_G[(\Pi_z \otimes I_F)\Gamma_{GF}(\Pi_z \otimes I_F)], \quad W_z = \mathrm{diag}(u_z, 1)H. \tag{31}$$

The outcome probability is independent of the payload and reference. The receiver measures the local spectral projectors as carriers arrive, stores their eigenphase product classically, and applies

$W_z^\dagger$ to the final flag. The result is exactly ρ_{RQ} on every branch. The anchor branch uses $V_0 = I$ and no correction.

For a Pauli word this means measuring each carrier in its indicated X, Y, Z basis, discarding identity factors, and retaining only the parity b of negative eigenvalues. The flag correction is HZ^b: first Z^b, then H. The receiver uses no inter-carrier quantum gate and no coherent delay for the gauge. At most one arriving carrier, a fixed meter ancilla and the output flag need be present. The source record, field arithmetic and timing counter use $O(\log n)$ classical bits, and the measured outcomes need only one parity bit.

The positive operation works for *any* gauge state independent of the payload/reference, including correlations between gauge carriers and hidden purifiers. The hard theorem uses the actual tracial gauge. No bath purifier, preparation label, source measurement or coherent return link is supplied. The record gives the *source encoding program*; the eigenvalues used for correction are measured from the actual emitted particles.

For a CNOT factor, even a two-qubit eigenmeasurement is unnecessary:

$$\mathrm{CNOT}\,|c\rangle_Z|b\rangle_X = (-1)^{cb}|c\rangle_Z|b\rangle_X. \tag{32}$$

Separate Z and X measurements and a Boolean AND provide its eigenvalue. Fixed-range matching factors need only a bounded number of pending classical outcomes, including wraparound pairs.

The strongest simple late-record architecture is to keep every gauge carrier and the flag. Once the record arrives, it performs these same local measurements and correction. This post-record readout uses $O(n)$ quantum operations rather than a large coherent inverse. The lower bound concerns its extensive retained memory, not an invented computational bottleneck. Known-basis environment-assisted correction is established [3]; the new theorem quantifies what the timing of the basis record costs in this payload task.

6 Physical source, timing, errors and qualification

6.1 An explicit one-flag Pauli source

The source holds the unknown payload flag and a classical branch/seed record. In the nonanchor branch it applies H to the flag, then interacts it once with each incoming carrier $I/2$ through the specified controlled Pauli. Each carrier leaves permanently after that encounter. Finally, a full SWAP transfers the flag to a fresh outgoing carrier and leaves the original source maximally mixed. The original source is discarded. The source needs one coherent payload qubit, one current carrier and $O(\log n)$ classical program storage.

The same record can be emitted through classical signalling, or through the same particle stream by swapping physical computational-basis control bits into fresh tracial carriers. The latter uses L extra carriers. In the early version, the source keeps a classical copy of the seed for its subsequent gate program; preparing and retaining that duplicate costs $O(L)$ classical bit writes and storage. These are explicit source-controller records, not bath purification keys. Both versions can use the same duplicated classical preparation. Record slots, source gates, carrier labels and arrival times are fixed and do not leak the selector before its declared disclosure.

For example, a universal local gate cell with calibrated CNOT, H, S, X and SWAP operations implements each controlled Pauli with at most five one- or two-qubit elementary gates, including negative-control conjugation. Source work is at most $5n + O(L^2 + 1)$ such gates with a conservative serial metadata routing layout. The receiver uses at most two basis rotations and one readout

per carrier, a parity update, and two final flag gates. All controls have fixed local arity and bounded norm. The source encounters here are strong finite gates; the proof does not claim small per-encounter action.

Peak and slew can both be finite. Write the Hamiltonian as $\hbar g(t)G$ with $\|G\| = 1$, so J and S below bound $|g|$ and $|\dot{g}|$. For area A (at most $\pi/2$ for the involution gates), a pulse with peak at most J and slope at most S uses a triangular envelope of duration $2\sqrt{A/S}$ if $A \le J^2/S$, or a trapezoid of duration $A/J + J/S$ otherwise. Its constant generator makes the area calculation exact. Switching and settling between available generators is separately charged. If this native set is unavailable, a supplied local compiler and its cost are required; bounded norm alone does not create arbitrary gates.

A circuit meter can reuse one ancillary qubit but must prepare/reset it for each local readout: $O(n)$ meter initializations, with their purity, time, energy and errors, belong in the receiver bill. A direct physical detector has corresponding reset and readout costs. Constant retained quantum memory does not mean constant cumulative preparation resources.

If a caller can instead command an immediate full SWAP of the initial source, that simpler interface exports the payload directly. Our comparison fixes the emitter's declared program and varies when its independent classical record is available. The theorem supplies no reason to build this encoder solely to transfer a qubit, and no claimed advantage over direct SWAP access. Its purpose is a sharp receiver-memory boundary for a physically specified mixed stream.

6.2 Execution bill

Let τ_g bound the duration of a compiled local gate, τ_m a readout, τ_b a metadata bit slot, and f_c the classical bit-operation rate. A serial sufficient schedule has

$$T = O\left(n(\tau_g + \tau_m + \tau_{\text{reset}} + \tau_{\text{route}} + \tau_{\text{settle}}) + L^2\tau_g + L\tau_b + \frac{nk^2}{f_c}\right). \tag{33}$$

Delivery flight times and any clock synchronisation are added. At fixed local hardware rates, quantum processing is linear and total classical work is near-linear. Precomputing the whole Pauli word removes online field arithmetic only by paying $O(n)$ program RAM and the same prior computation.

The source flag must remain coherent for the entire run. A fixed logical source dimension is not a claim of arbitrarily long protection with fixed noisy hardware. The early receiver's coherent carrier storage is constant in size and $O(n)$ in total qubit-time at fixed gate/readout cadence. Direct late storage has $n+1$ coherent qubits and $O(n^2)$ qubit-time. Section 3.1 proves the matching quadratic lower bound in expected qubit-time under the maximally mixed/Bell-marginal input, even with variable classical branch sizes. If serial field generation lengthens slots, both clocks acquire its paid factor. Loading/routing the late bank is additional; its post-record recovery is linear.

The tracial hard-source factory must produce the stated independent state. For independent qubits with Bloch lengths β_i, the inherited sufficient bound is

$$\nu_{\text{init}} \le \tfrac{1}{2}\sqrt{\prod_i (1+\beta_i^2) - 1}. \tag{34}$$

Thus uniform $\beta_i = O(\nu/\sqrt{n})$ is sufficient for total initialization error ν. Ideal independent Pauli twirling with erased/inaccessible keys is one paid preparation option. Approximate twirls of

correlated raw inputs need composable channel bounds. The positive decoder's exact instrument identity does not itself require a tracial factory, but the quantitative hard-source memory claim does.

6.3 Composable errors and what calibration can establish

A sufficient positive-reader error bound is

$$\epsilon_{\text{actual}} \le \nu_{\text{prep}} + \nu_{\text{record}} + \nu_{\text{loss}} + \sum_{\ell=1}^{G} \delta_\ell + \nu_{\text{storage}} + \nu_{\text{timing}}. \tag{35}$$

Each term is a composable half-diamond or outcome-error bound for the actual history/context in which it is used. A simple uniform allocation requires $\delta_\ell = O(\epsilon/G) = O(\epsilon/n)$. Classical parity readout errors also accumulate; fixed raw local error is not a constant-error large-n theorem. For Hamiltonian-norm bound J_E, timing error contributes at most $J_E|\Delta t|/\hbar$ per affected pulse. A declared source memory-noise rate contributes its rate times the full source exposure. Correlated memory noise needs a bound for the joint process, not an inferred scalar coherence time. Fault-tolerant control, if used, has additional physical qubits, cycles, initialization and calibration costs; none is silently supplied.

If the actual flagged complete emission differs from the ideal by half diamond ν, under the same fixed emission/record schedule, a receiver with actual error ϵ recovers the ideal emission with error at most $\epsilon + \nu$. The capacity bound therefore uses that sum, assumed at most $1/2$. The five-bit expander version below is particularly demanding.

Expected occupancy additionally depends on branch probabilities. Define $f_\ell(r) = \max\{0, \ell(r - 2^{-\ell})/(1 - 2^{-\ell})\}$ as in (16). For the same receiver instrument, branch-to-capacity assignment and physical cut in both experiments,

$$\mathbb{E}_{\text{actual}} q_y \ge \max\{0, f_\ell(r_{\epsilon+\nu}) - \ell\nu\}. \tag{36}$$

The ideal rank/chord proof bounds the expectation of $z_y = \min(q_y, \ell) \in [0, \ell]$. Data processing bounds the two prefix record laws' total variation by ν, changing this bounded expectation by at most $\ell\nu$; also $q_y \ge z_y$. Integrating (36) gives the actual exposure bound. Fixed timing and absence of extra side channels are essential: final-channel closeness alone does not control earlier metadata arrival or arbitrary changes in the clock.

Generic qualification of a constant number of stationary local gate types to $O(\epsilon/n)$ by tomography costs conservatively $O(n^2\epsilon^{-2}\log(1/\delta))$ trials. This is an upfront cost, not linear online processing. End-to-end qualification is also possible: for the random selector distribution the complete early reader is a qubit channel, so trusted two-qubit Choi tomography uses $O(\zeta^{-2}\log(1/\delta))$ runs for additive half-diamond tolerance ζ, each costing the full time (33). Measure the 15 nonidentity two-qubit Paulis to accuracy h; then $\left\|J - \widehat{J}\right\|_1 \le \sqrt{15}\,h$ and $d_\diamond(\Lambda, \mathrm{id}) \le \left\|\widehat{J} - \Phi_2\right\|_1 + \sqrt{15}\,h$. Trusted Bell preparation, protection of its reference throughout the run, readout, selector sampling and stationarity are required. This qualifies an operation that performs well; it does not improve an inaccurate operation. It certifies the averaged channel, not every seed separately. A worst-seed claim requires additional structure or a correspondingly larger qualification.

7 A constant-selector, fixed-register source

The logarithmic-record construction has useful robustness and one coherent source qubit, but its classical state grows with $\log n$. For a separate fixed-register version, import the explicit 16-generator theorem of Anshu et al. [7], already accepted in the warm expander companion. It gives Hermitian disjoint Pauli/CNOT layers on $n = 3s$ qubits, ring range at most four, with absolute gap $\gamma_0 = 1/(2 \cdot 10^8)$.

Use the 16 operators $X \otimes U_j$ and anchor $Z \otimes I$, uniformly. Equation (21) gives

$$\kappa_0 = \frac{32\gamma_0}{17} = \frac{16}{17 \cdot 10^8} \simeq 9.4117647 \times 10^{-9}. \tag{37}$$

There are 17 labels and a five-bit record. At $\epsilon \le \kappa_0/16 \simeq 5.88235 \times 10^{-10}$, the late reader needs capacity for at least n qubits. This threshold is fixed independently of n but too small for a one-percent claim. The small-bias gap cannot be assigned to this source.

The full generator list, lift proof and routing schedule are supplied in the local-expander implementation companion. Hold the first four ring sites and use a four-site look-ahead window. Emit in the common order $5, 6, \dots, n, 1, 2, 3, 4$, then flag and record. Every nonwrapping pair has separation at most four; every wrapping partner lies in the retained boundary. Ten source qubits suffice for boundary, window, current carrier and flag. Controlled CNOTs are Toffolis with constant local compilation cost. Pad each slot to the worst-case duration so labels and timing do not disclose j. Small cases use the same finite buffer.

This is a time-programmed finite source with $O(n)$ instructions and an explicitly paid external clock/horizon, not an immutable time-homogeneous thermal generator. An internal position counter costs $O(\log n)$ classical bits if that timing service is removed. The flag and boundary carriers need protection for $O(n)$ time; a fixed logical register does not make this free.

With early j, individual control-Z/target-X measurements give every CNOT eigenvalue. At most four pending classical outcomes plus parity are needed; no coherent periodic buffer is required. With late j, the main theorem and every-prefix bound apply even to collective compression. The selector can alternatively be an arbitrary constant-dimensional quantum input emitted last: retaining the outputs makes the controlled encoding invertible, and classical-selector subcases prove the lower bound. The early protocol instead assumes a classical selector and does not preserve its unknown quantum coherence. In that variant the five selector qubits also count as coherent source storage, and their controlled local gates need their own constant-size compilation.

Construction	Record bits	Certified κ	Paid source storage
Uniform Pauli	$2n + 1$	$4/3$	One coherent flag; $O(n)$ classical bits
Small-bias Pauli	$2\lceil \log_2(4n - 2) \rceil + 1$	$4/5$	One coherent flag; $O(\log n)$ classical bits
Local expander	5	$16/(17 \cdot 10^8)$	Ten coherent registers; fixed selector; external time program

Table 1: Complete but different constructions. Their source, record and gap resources cannot be interchanged. A current carrier and meter add fixed workspace.

8 Comparison with the strongest applicable approaches

Ballester, Wehner and Winter already show that one Boolean answer after a late basis choice can require full quantum memory for perfect prediction [1, Theorem 6.6]; their discussion after that construction raises the near-perfect case. The bare late-choice phenomenon is therefore not new. Bluhm, Rauber and Wolf already allow unrestricted classical side records and prove exact measurement-relative incompressibility and fixed-family small-error stability [2, Theorem 5.1]. Their compactness proof does not supply a dimension-uniform numerical threshold. Their common decompressor preserves specified observables for all input states.

Ioannou et al. [4] provide the closer operational framework: one quantum/classical compression followed by setting-dependent measurements, with a low-rank Kraus characterization. Their analytic robustness bounds for the full unitarily symmetric family of noisy rank-one projective measurements are substantive prior results. That all-measurement premise does not follow from our finite family of balanced binary probes and its average score, so those bounds do not directly replace the gap-dependent implication here. Setting-dependent late measurements themselves are not a new model.

The lower bound uses only average discrimination of the finite binary probes; it does not require preservation of their statistics on every input. Each selector may choose a different decoder returning only one encoded qubit. Equation (9) connects this to one analytic reconstruction, and the gap yields the explicit dimension bound. The arbitrary-reference encoding, physical stream, and causal prefix argument complete the operation and its memory-exposure cost. Petz recovery, commutator rigidity and Schmidt-number bounds remain existing machinery.

The early protocol is an instance of environment-assisted correction [3]. An equally informed early-record receiver can use the same local measurements; no advantage over that baseline is claimed. Classical small-bias sets and their Pauli use are prior machinery [5, 6], and the five-bit construction imports the local expander [7]. We do not claim better expanders, encryption, or a first measurement-relative compression theorem.

Pure-input coherent absorbers and collectors use a different physical resource: actual pure input ports [8]. An inaccessible purifier cannot be substituted for an observed output. Small matrix-product bond dimension alone does not construct a CPTP compression. The present hard family strengthens a whole-output obstruction to one arbitrary payload, but is not the homogeneous weak partial-SWAP channel. Its specialised small-memory receiver remains open.

9 Executed evidence and reproducibility

The package contains complete proofs, programs, machine-readable receipts and the independent audits. These audits are by separately tasked collaborating agents, not external peer review or a proof-assistant certificate. The following are executed finite calculations, not hardware demonstrations or numerical replacements for the theorem.

- The actual local-measurement decoder was checked for all Pauli selectors and anchors for $n = 1, 2, 3, 4$: 344 cases. Its full flag-plus-gauge input Choi map equals partial trace, with maximum normalised-Choi half-trace residual 3.89×10^{-16}. Correlated mixed GHZ gauge checks give residual at most 9.99×10^{-16}.
- Direct commutator superoperators for the uniform Pauli family at $n = 1, 2, 3$ have a one-dimensional kernel and gap $4/3$, to numerical precision. Thirty-two hybrid compression examples, including nonunital amplitude damping, reset, classical block records and singular baselines, obey

$q_j \geq t_j^2$, $F_e \leq m/N$ and the gap bound. Maximum trace-preservation residue is 3.31×10^{-15}. Two examples saturate the latter rank and gap steps.

- Measuring one gauge qubit in Z before the record, retaining that classical result, and then applying the specified late decoder gives $(5/6)\,\mathrm{id} + (1/6)\,\mathrm{Ad}_X$ for the uniform Pauli-plus-anchor family. Checks at $n = 1, 2, 3$ have Choi residual at most 3.92×10^{-16}; its half-diamond error is analytically $1/6$. This is a baseline, not an SDP optimum.
- Exact finite-field and character checks were run for the small-bias construction at $n = 1, 2, 3, 4, 5, 8$. For $n = 1000$, the field polynomial $x^{12} + x^3 + 1$ was found after five candidate tests, and one complete 1000-site Pauli program was generated. No 1000-qubit density matrix was simulated.

The Choi residuals above are not reported as numerically evaluated diamond norms. The analytic identities establish the reference-uniform statements. Large source factories, calibration, bounded-noise hardware and protected memories were not simulated.

10 Independent completed companions

10.1 Restricted partial-SWAP receiver obstruction

A source collides with fresh $I/2$ qubits at angle $0 < \theta < \pi/2$. A single tracial receiver qubit then uses an arbitrary predetermined partial-SWAP angle on each outgoing particle and discards it. With $c = \cos\theta$, $s = \sin\theta$, its depolarizing visibility is uniformly bounded by

$$|B_N| \leq L(\theta), \qquad L(\theta)^2 = \frac{s^2}{1+c^2} + \frac{32 s^4}{(3-c)^2}. \tag{38}$$

An arbitrary final decoder has half-diamond error at least $\frac{3}{4}\max\{0, 1 - L(\theta)\}$. At $\theta = 0.1$ this is 0.6928864. The proof uses a closed three-dimensional operator recurrence; 103 direct 8×8 checks have maximum discrepancy 3.34×10^{-16}. It excludes this scalar receiver, not pure or larger memories, general instruments, measurement feedback, subsystem codes or block compression. No positive general homogeneous weak-stream receiver was obtained.

10.2 Finite thermal intervention for hidden linear dynamics

For a certified observable all-even OU system $dX = AX\,dt + \sqrt{2D}\,dW$, $Y = CX$, with A Hurwitz and $D > 0$, a common thermal port scaling all diffusion by $1 + u(t)$ while preserving A supplies new information absent from the warm force-only interface. A raised-cosine pulse $u(t) = \eta \sin^2(\pi t/h)$ has an invertible covariance transfer

$$\mathcal{T}_A = \eta\omega^2 (e^{h\mathcal{L}_A} - I)[\mathcal{L}_A(\mathcal{L}_A^2 + \omega^2 I)]^{-1}, \quad \mathcal{L}_A(Z) = AZ + ZA^T, \quad \omega = 2\pi/h. \tag{39}$$

Random hot/control assignment gives a martingale-unbiased difference of post-pulse covariance matrices on one continuously running specimen, without hidden resets. Shifted tagged moments recover unknown A, D up to similarity and the model's native entropy production, with declared order, observability, anti-aliasing, diffusion-positivity and thermal-reach premises. Inversion uses a Stein solve and Lyapunov-map applications, $O(d^3)$ classical arithmetic.

The companion pays exposure, moments, conditioning, finite bins, digitisation, temperature slew, calibration and drift errors. A million no-reset cycles were executed on a two-dimensional model: inferred entropy rate 21.07725 versus 20.25. Its conservative confidence radius is too loose to certify that displayed error. General covariance identification is a strong baseline [9]; no efficiency superiority

or arbitrary hidden-dissipation coverage is claimed. Thermodynamic entropy interpretation requires the declared physical state variables, parity and bath model; unseen extra modes remain outside it.

10.3 Translational force from raw blurred images

The other independent construction uses a Stokes virtual field and one acquired radial point-spread second moment. Its quadratic displacement kernel is separately biharmonic/harmonic, so a finite Laplacian correction cancels the specified radial blur exactly. Five weighted image covariances and a positive normalizer give a translational-force current; a reciprocal identity supplies the force with paid boundary, forcing and finite-lag terms. An explicit annular translation field and manufactured-flow checks close the mechanical operation.

This requires the stated optical-volume access, independent particle-pair law, PSF shape/support and moment calibration, geometry, and positive normalization. The direct estimator can cost $O(h^{-4})$ records as lag h shrinks. The executed reciprocal force 1.9000352368911049 agrees with direct wall stress 1.9000352368911066; these are deterministic quadrature, not measured images. Pizzetti/Faxen corrections, reciprocal-force formulas and correlation-based stress inference are strong prior machinery. An equally informed implementation can compile the same weighted covariances. No general advantage over a direct force balance or the best image-correlation estimator is established.

11 Preserved knowledge and unresolved boundaries

The warm results are retained unchanged, with governing sources and access contracts. They are premises, not subjects of another validation campaign.

1. **Fourth-order action law.** Fresh finite tracial ancillas are independent of source/reference and retained systems conditional on genuine classical history. No coherent return or extra source measurement/reset is supplied. With whole-encounter actions a_j, $b = \prod_j [1 - \min(1, 4a_j^4)]$, $\alpha = 1 - b$ and $c = 1 - d^{-2}$, every decoder has error at least $c(1 - \sqrt{\alpha}) \geq cb/2$, with the accepted acquired-information bound. Classical feedback and finite stopping are included under that contract. Pathwise charge $\sum_j -\log[1-q(a_{j,h})] \leq C_*$ gives $b = e^{-C_*}$; this is not an expected-action or free-postselection result.
2. **Exact partial-SWAP channels.** The acquired channel is $\mathcal{C}_N(\rho) = \mathrm{Tr}_S\, U(\rho \otimes I_B/d^N)U^\dagger$. Reversing the circuit on all outputs and an independent tracial dummy gives $\mathcal{D}_N\mathcal{C}_N = (1 - r)\,\mathrm{id} + r\mathcal{R}_{I/d}$, with $r = \prod_j (1 - \sin^4\theta_j)$. Original source and purifiers are absent. The explicit half-diamond error is cr and every decoder has error at least $c(1 - \sqrt{1-r})$, the factor-two guarantee. This is an exact channel identity, not exact finite-weak-collision recovery.
3. **Independent modules.** Product emission allows arbitrary entangled inputs/references and collective final decoding. For $V = \prod_i (1 - c_i r_i)$, the explicit reader has error $1 - V$ and every collective decoder error at least $1 - \sqrt{V}$, with the sharper accepted finite-dimension version available. Active coupling that destroys product emission is outside the theorem.
4. **Forgetting and acquisition.** Source visibility $\prod_j \cos^2\theta_j$ decays on the quadratic collision scale while output recovery uses the quartic scale. Identical reduced source dynamics can have different physical acquired channels. Factory, routing, calibration, protection, peak and slew costs remain part of those results.
5. **Earlier families.** Unknown-support rare-failure acquisition, the fixed-start lifetime bound, correlated product-invariant computation, finite collision/factory implementation and separate

equilibrium-safe initialization retain their operating refresh backbone, declared local port reach, starting-mass and precision costs. The all-strength hidden-dephasing obstruction, finite effusion protocol, protected readout and pure-Weyl-port recovery retain their supplied physical resources.

6. **Both prior nonquantum companions.** The force-port OU nonidentifiability construction and directed-local-dynamics prediction obstacle remain at their original scopes. The latter uses an ideal local compiler and is not a fault-tolerant physical complexity separation. The new thermal pulse adds a different physical port and does not invalidate the earlier identical-transcript obstruction.
7. **Pure-port weak export.** The earlier effort to remove the strong-export threshold did not produce an efficient finite decoder. Forgetting and abstract recovery existence did not close overlapping output recovery and boundary correlations. This remains a construction gap, not a negative theorem. The completed pure coherent Weyl collector remains an available stronger-export baseline with growing control/precision and paid protected memory.

The homogeneous weak partial-SWAP stream's small-memory, near-linear causal receiver also remains unresolved. The present result changes the general question: there is no universal small-memory inference from payload dimension alone, and in the programmed-source class from source-register dimension alone. A short early control record supplies an exact constant-workspace reader in the constructed family. The same record delivered late costs extensive memory and quadratic retained quantum exposure. These conclusions and their stated limits are the completed scientific consequence of this return.

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